

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech. Branch: Civil Engineering Semester: VII

Subject Code & Name: Design of Concrete Structures – II (BTCVC701_Y20)

Max Marks: 60 Date: 05/02/2025 Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.

2. Question No. 1 will be compulsory and include objective-type questions.

3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.

4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.

5. Use of non-programmable scientific calculators is allowed.

6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Torsion gives rise to	CO1	1
	a. Only Additional Moment b. Only Additional Shear Force c. Both Additional Moment and Shear Force d. None of the above		
2	When $M_T > M_U$	CO1	1
	a. $M_{e2}=0$ b. $M_{e2}=M_T-M_U$ c. $M_{e2}=M_T+M_U$ d. $M_{e2}=M_T/M_U$		
3	In case of torsion, the side face reinforcement is provided if the overall depth of beam exceeds	CO1	1
	a. 450 mm b. 600 mm c. 750 mm d. 900 mm		
4	The area of side face reinforcement required is	CO1	1
	a. 0.1% A_g b. 0.2 % A_g c. 0.4 % A_g d. 0.8% A_g		
5	Axially loaded column is subjected to	CO2	1
	a. Only Axial Force b. Axial Force and Moment in one direction c. Axial Force and Moments in two directions d. None of the above		
6	Uni-axially loaded column is subjected to	CO2	1
	a. Only Axial Force b. Axial Force and Moment in one direction c. Axial Force and Moments in two directions d. None of the above		
7	Condition of short column is	CO2	1

	a. Slenderness ratio ≤ 12 along X axis	b. Slenderness ratio ≤ 12 along Y axis	c. Slenderness ratio ≤ 12 along both X and Y Axis	d. None of the above		
8	Minimum area of steel recommended in column as per IS 456:2000 is				CO2	1
	a. 0.8 % Ag	b. 1.0 % Ag	c. 0.6 % Ag	d. 1.2 % Ag		
9	The process of tensioning the cable after the setting of concrete is called as				CO3	1
	a. Pre-stressing	b. Pre-tensioning	c. Post-tensioning	d. None of the above		
10	Which of the following is not an immediate loss				CO3	1
	a. Loss due to elastic shortening	b. Loss due to anchorage slip	c. Loss due to friction	d. Loss due to creep		
11	In case of loss due to friction with jacking at both ends, maximum loss is observed at				CO3	1
	a. Jacking end	b. Unjacked end	c. Mid-section of the beam	d. None of the above		
12	Which of the following is the system of post tensioning				CO3	1
	a. Freyssinet Anchorage System	b. Gifford Udall System	c. Magnel Blaton System	d. All of the above		
Q. 2	Solve the following.					12
A)	An RC beam (350 x 800) mm is subjected to the factored bending moment of 215 kN-m, factored shear force of 150 kN and factored torsion of 105 kN-m respectively. Assume M30 grade concrete and Fe415 grade steel, design the reinforcement for the beam. Take effective cover as 50 mm.				CO1	6
B)	Design a rectangular beam section of size 250 x 450 mm subjected to service bending moment of 30 kN-m, service shear force of 40 kN, service torsion moment of 20 kN-m. Use M20 grade concrete and Fe415 grade steel. Take effective cover as 40 mm.				CO1	6
Q.3	Solve the following.					12
A)	Design the reinforcement in a column of size 450 mm x 600 mm, subject to an axial load of 2000 kN under service dead and live loads. The column has an unsupported length of 3.0 m and is braced against side sway in both directions also both ends are rotationally fixed. Use M20 concrete and				CO2	6

	Fe415 steel.		
B)	Design the helical reinforcement in a circular column subjected to a factored load of 1500kN. The column has an unsupported length of 3.2 m and is braced against side sway. Use M25 concrete and Fe415 steel.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Compare pre-tensioning and post-tensioning.	CO3	6
B)	Explain the various types of pre-stressing.	CO3	6
C)	Explain in brief Magnal-Blaton system of pre-stressing.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	A rectangular concrete beam 100 mm wide and 250 mm deep spanning over 8 m is prestressed by a straight cable carrying an effective prestressing force of 250 kN located at an eccentricity of 40 mm. The beam supports a live load of 1.2 kN/m. Calculate the resultant stress distribution for the centre of span cross section of the beam assuming density of concrete as 24 kN/m ³ .	CO4	6
B)	A symmetrical I-Section beam is used to support an imposed load of 2 kN/m over a span of 8m. The sectional details are as, flange width is 300 mm, flange thickness is 60 mm, web thickness is 80 mm and overall depth of the beam is 400 mm. At the center of span, the effective prestressing force of 100 kN acts at a distance of 50 mm from the soffit. Estimate the stress at the center of span for the combined effect of prestress, dead load and live load.	CO4	6
C)	A prestressed concrete beam with a rectangular section 120 mm wide and 300 mm deep supports a UDL of 4 kN/m which includes the self-weight. The effective span of the beam is 6 m. The beam is concentrically prestressed by a cable carrying a force of 180 kN. Locate the location of pressure line in the beam.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the various types of losses in prestressing.	CO3	6
B)	A pretensioned beam 200 mm wide and 300 mm deep is prestressed by 10 wires of 7 mm diameter initially stressed to 1200 N/mm ² , with their centroids located 100 mm from the soffit. Find the maximum stress in	CO4	6

	concrete immediately after transfer, allowing only for elastic shortening of concrete. If the concrete undergoes a further shortening due to creep and shrinkage while there is a relaxation of five percent of steel stress, estimate the final percentage of loss of steel stress. Take E_s as 210 kN/mm^2 , E_c as $5700\sqrt{f_{ck}}$, f_{ck} as 42 N/mm^2 , Creep coefficient (ϕ) as 1.6 and total residual shrinkage strain as 3×10^{-4} .		
C)	A post tensioned prestressed concrete beam of rectangular cross section 150 mm wide to be designed for an imposed load of 3 kN/m over a simply supported span of 7.5 m . The stress in concrete must not exceed 10 kN/mm^2 in compression and 1.5 kN/mm^2 in tension at any time. Loss in prestress is 14%, calculate the minimum depth required. Also find the number of cables required if the prestress is 1250 N/mm^2 . Assume diameter of cable as 5 mm .	64	6
*** End ***			